# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – Debugging & Optimization Task

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| | | | |
|---|---|---|---|
| **Course Code:** | CSA0309 | **Course Title:** | Data Structures |
| **CO Assessed:** | CO2 – Manipulation (BL3, BL4) | **Assessment:** | Assessment Tool 2 – Debugging & Optimisation Task |
| **Weightage:** | 35% of CIA | **Total Marks:** | 100 |
| **CO2 Statement:** | *Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.* | | |
| **Student Name:** | Kanish Kumar V | **Reg. No.:** | 192524361 |
| **Section / Batch:** | B | **Date:** | 22/07/2026 |

### Code of Conduct

*I Kanish Kumar V / 192524361 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Kanish Kumar V

### Instructions

- This test contains 20 Debugging & Optimization Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic | Focus | Q Nos | Marks |
|---|---|---|---|
| **Search Data Structure, Stack** | Binary Search, Stack Operations | 1 - 5 | 25 |
| **Linked List Data Structure** | List Operations | 6 – 10 | 25 |
| **Queue Data Structures** | Queue Operations | 11 - 15 | 25 |
| **Queue Data Structures, Stack Notations** | Queue Operations, Expression Evaluation | 16 - 20 | 25 |
| | **GRAND TOTAL:** | **100 Marks** | |

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Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

else
high = mid;
}
return -1;
}

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;

    while (low <= high) {
        int mid = low + (high - low) / 2;

        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }

    return -1;
}
```

Test Input

```
arr = {2, 4, 6, 8, 10};
key = 10;
```

Output

```
Compilation Error:
C:/CCPPComp/CCPP2/bin/../lib/gcc
/x86_64-w64-
mingw32/10.3.0/../../../../x86_6
4-w64-mingw32/bin/ld.exe:
```

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Submitted

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Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Find the error in the stack deletion operation. Justify it.
int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int pop() {
    if (top == -1) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_19252436
1RB_6a74b3d18c8281.80517771\main
.c: In function 'pop':
C:\Windows\TEMP\sandbox_19252436
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List Data Structures 5 marks

## Q3 Stack

5 marks

Check the snippet and identify the logical error.

int stack[5];

int top;

void init() {

top = 0;

}

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int stack[5];
int top;

void init() {
    top = -1;
}
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:/CCPPComp/CCPP2/bin/../lib/gcc
/x86_64-w64-
mingw32/10.3.0/../../../../x86_6
4-w64-mingw32/bin/ld.exe:
C:/CCPPComp/CCPP2/bin/../lib/gcc
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List Data Structures 5 marks

## Q4 Stack

5 marks

Identify the errors in the following snippet.

if(ch == ')') {

while(stack[top] != ')') {

postfix[j++] = pop();

}

pop();

}

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if(ch == ')') {
    while(stack[top] != '(') {
        postfix[j++] = pop();
    }
    pop();
}
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_19252436
1RB_6a74b485c0ce71.37640441\main
.c:1:1: error: expected
identifier or '(' before 'if'
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List Data Structures 5 marks

## Q5 Stack

5 marks

In balancing Bracket, you identify the error.
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
pop();
} else {
return 0;
}

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if (top < 0) return 0;

if ( (ch == ']' && stack[top] == '[') ||
     (ch == '}' && stack[top] == '{') ||
     (ch == ')' && stack[top] == '(') ) {
    top--; // pop
} else {
    return 0;
}
```

Test Input

stdin input (optional)

Output

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List Data Structures 5 marks

## Q6 Linked List

5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
temp = head
while temp is not None:
    print(temp.data)
    temp = temp.next
```

Test Input

stdin input (optional)

Output

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List Data Structures 5 marks

## Q7 Stack

5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
def POP(stack):
    if top == -1:
        print("Underflow")
        return None
    else:
        item = stack[top]
        top -= 1
        return item
```

Test Input

stdin input (optional)

Output

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks
- Q8

## Q8 Linked List Data Structures

5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
prev = None
while current is not None:
    nxt = current.next
    current.next = prev
    prev = current
    current = nxt
```

Test Input

stdin input (optional)

Output

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## CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks

Q9 **Linked List** 5 marks

What is the error in the below code?
new.next = head
head = new

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
new->next = head;
head = new;
```

Test Input

stdin input (optional)

Output

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- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List Data Structures 5 marks
- Q9

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
DEQUEUE(Q):
if front == -1:
    print("Underflow")
    return
else:
    x = Q[front]
    front = front + 1
    if front > rear:
        front = -1
        rear = -1
    return x
```

Test Input

stdin input (optional)

Output

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## CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- **Q1** Stack Notations - Postfix Expression 5 marks
- **Q2** Circular Queue 5 marks
- **Q3** Binary Search 5 marks
- **Q4** Queue 5 marks
- **Q5** Queue 5 marks
- **Q6** Doubly Linked List 5 marks
- **Q7** Stack 5 marks

**Q1 Stack Notations - Postfix Expression** 5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}
```

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int i = 0;
while (expression[i] != '\0') {
    if (isdigit((unsigned char)expression[i])) {
        push(expression[i] - '0');
    } else {
        int op1 = pop();   // right operand
        int op2 = pop();   // left operand
```

Test Input

stdin input (optional)

Output

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- **Q1** Stack Notations - Postfix Expression 5 marks
- **Q2** Circular Queue 5 marks
- **Q3** Binary Search 5 marks
- **Q4** Queue 5 marks
- **Q5** Queue 5 marks
- **Q6** Doubly Linked List 5 marks
- **Q7** Stack 5 marks
- **Q8** Stack 5 marks
- **Q9**

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}
```

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int dequeue() {
    if (front == -1) return -1;   // queue empty

    int data = queue[front];

    // move front forward circularly
    if (front == rear) {            // last element removed
        front = -1;
        rear = -1;
    } else {
        front = (front + 1) % MAX;
    }

    return data;
}
```

Test Input

stdin input (optional)

Output

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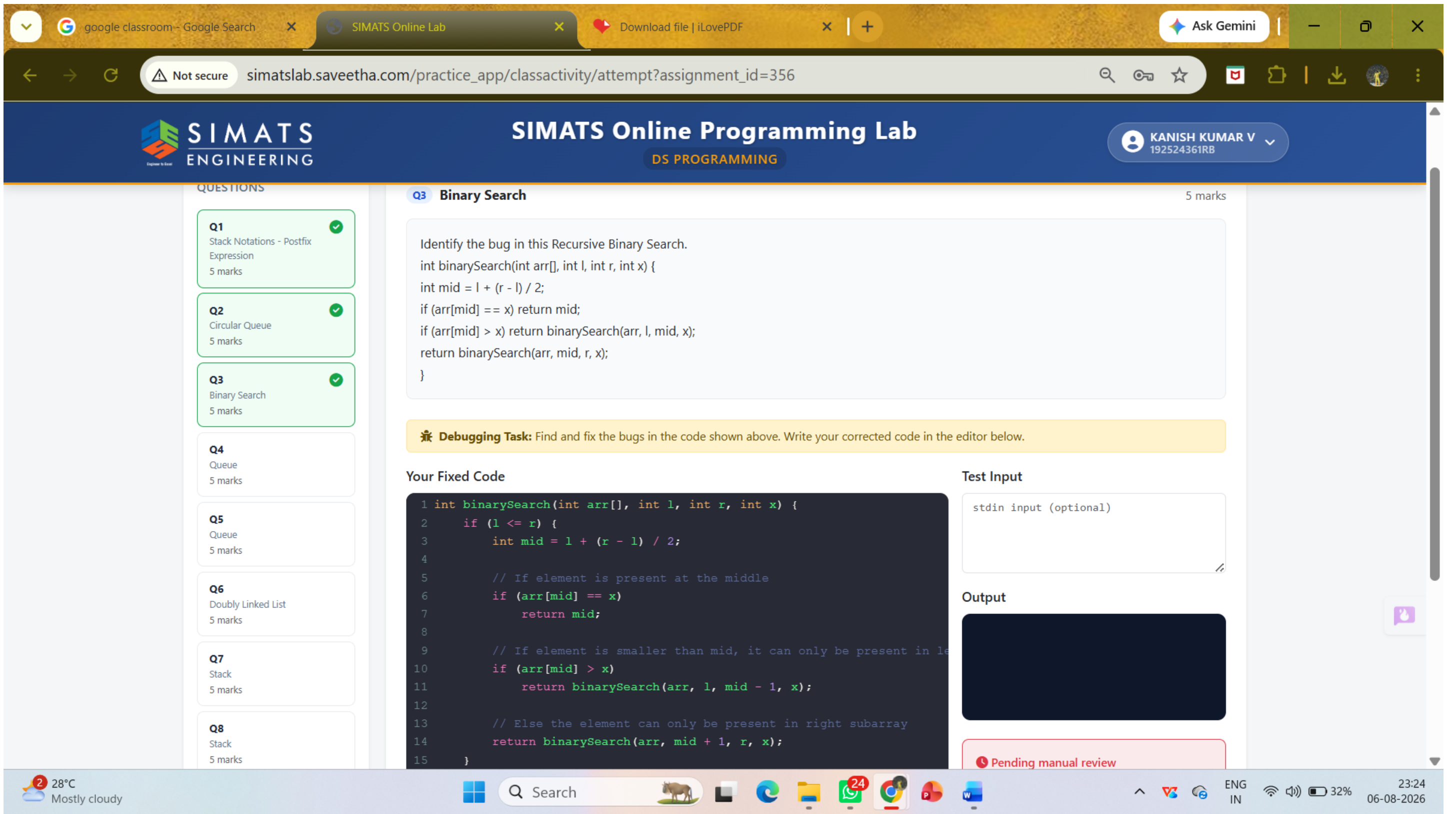

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QUESTIONS
Q1
Stack Notations - Postfix Expression
5 marks
Q2
Circular Queue
5 marks
Q3
Binary Search
5 marks
Q4
Queue
5 marks
Q5
Queue
5 marks
Q6
Doubly Linked List
5 marks
Q7
Stack
5 marks
Q8
Stack
5 marks
Q3 Binary Search
5 marks
Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid, r, x);
}
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
int binarySearch(int arr[], int l, int r, int x) {
    if (l <= r) {
        int mid = l + (r - l) / 2;
        // If element is present at the middle
        if (arr[mid] == x)
            return mid;
        // If element is smaller than mid, it can only be present in le
        if (arr[mid] > x)
            return binarySearch(arr, l, mid - 1, x);
        // Else the element can only be present in right subarray
        return binarySearch(arr, mid + 1, r, x);
    }
Test Input
stdin input (optional)
Output
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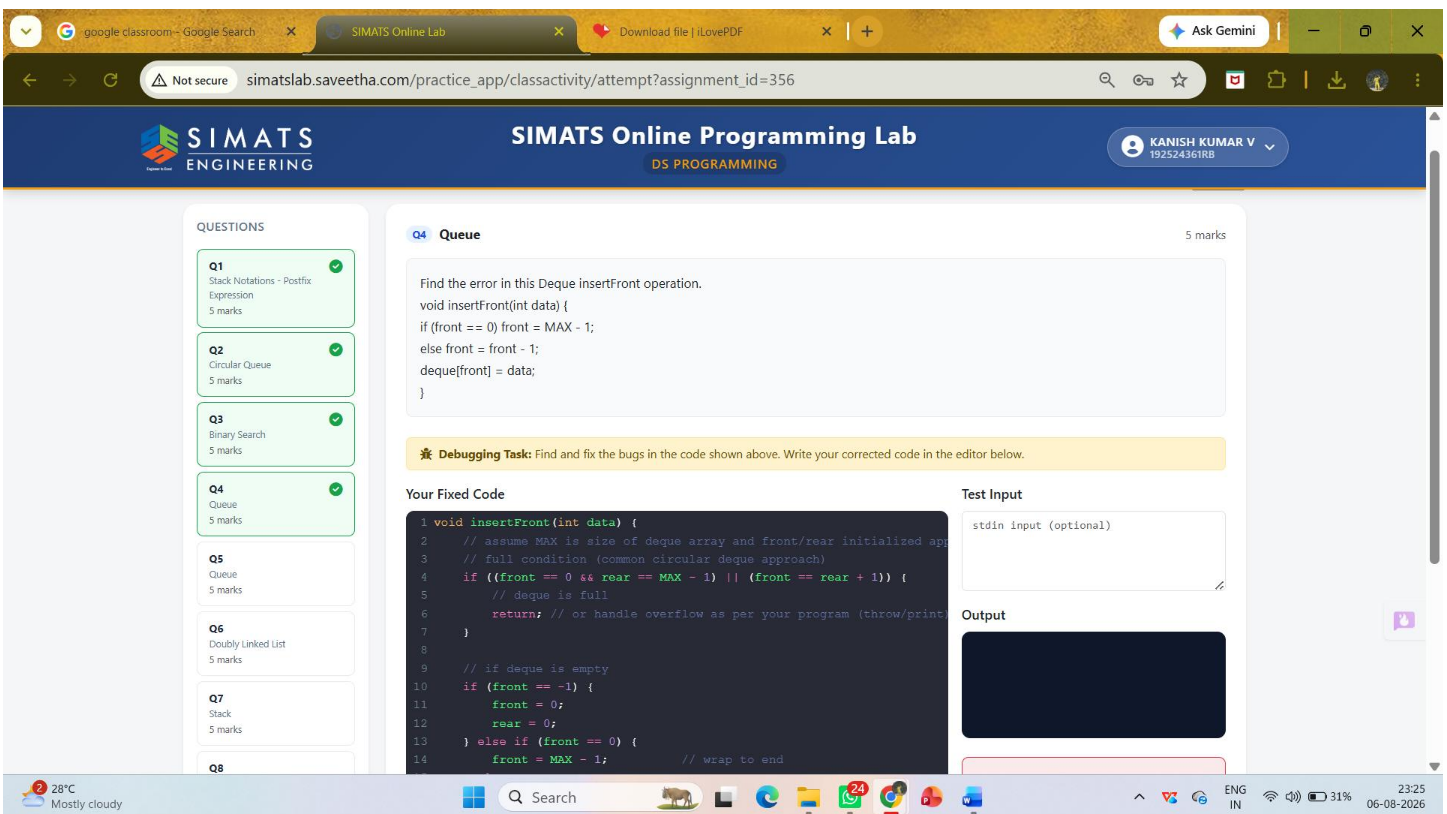

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QUESTIONS
Q1
Stack Notations - Postfix Expression
5 marks
Q2
Circular Queue
5 marks
Q3
Binary Search
5 marks
Q4
Queue
5 marks
Q5
Queue
5 marks
Q6
Doubly Linked List
5 marks
Q7
Stack
5 marks
Q8
Q4 Queue
5 marks
Find the error in this Deque insertFront operation.
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
void insertFront(int data) {
    // assume MAX is size of deque array and front/rear initialized app
    // full condition (common circular deque approach)
    if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
        // deque is full
        return; // or handle overflow as per your program (throw/print)
    }
    // if deque is empty
    if (front == -1) {
        front = 0;
        rear = 0;
    } else if (front == 0) {
        front = MAX - 1;           // wrap to end
Test Input
stdin input (optional)
Output
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## CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack Notations - Postfix Expression 5 marks
- Q2 Circular Queue 5 marks
- Q3 Binary Search 5 marks
- Q4 Queue 5 marks
- Q5 Queue 5 marks
- Q6 Doubly Linked List 5 marks
- Q7 Stack 5 marks

Q5 **Queue** 5 marks

Identify the error in the Queue enqueue logic.
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void enqueue(int data) {
    if (rear == MAX) return;    // queue full
    queue[rear] = data;         // insert at current rear position
    rear++;
}
```

Test Input

stdin input (optional)

Output

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QUESTIONS

- Q1 Stack Notations - Postfix Expression 5 marks
- Q2 Circular Queue 5 marks
- Q3 Binary Search 5 marks
- Q4 Queue 5 marks
- Q5 Queue 5 marks
- Q6 Doubly Linked List 5 marks
- Q7 Stack 5 marks
- Q8

Q6 **Doubly Linked List** 5 marks

Identify the error in this Doubly Linked List node deletion.
void deleteNode(Node* del) {
if (del->prev != NULL)
del->prev->next = del->next;
if (del->next != NULL)
del->next->prev = del->prev;
}

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void deleteNode(Node* del) {
    if (!del) return;


    if (del->prev == NULL) {
        head = del->next;
    } else {
        del->prev->next = del->next;
    }


    if (del->next == NULL) {
        tail = del->prev;
```

Test Input

stdin input (optional)

Output

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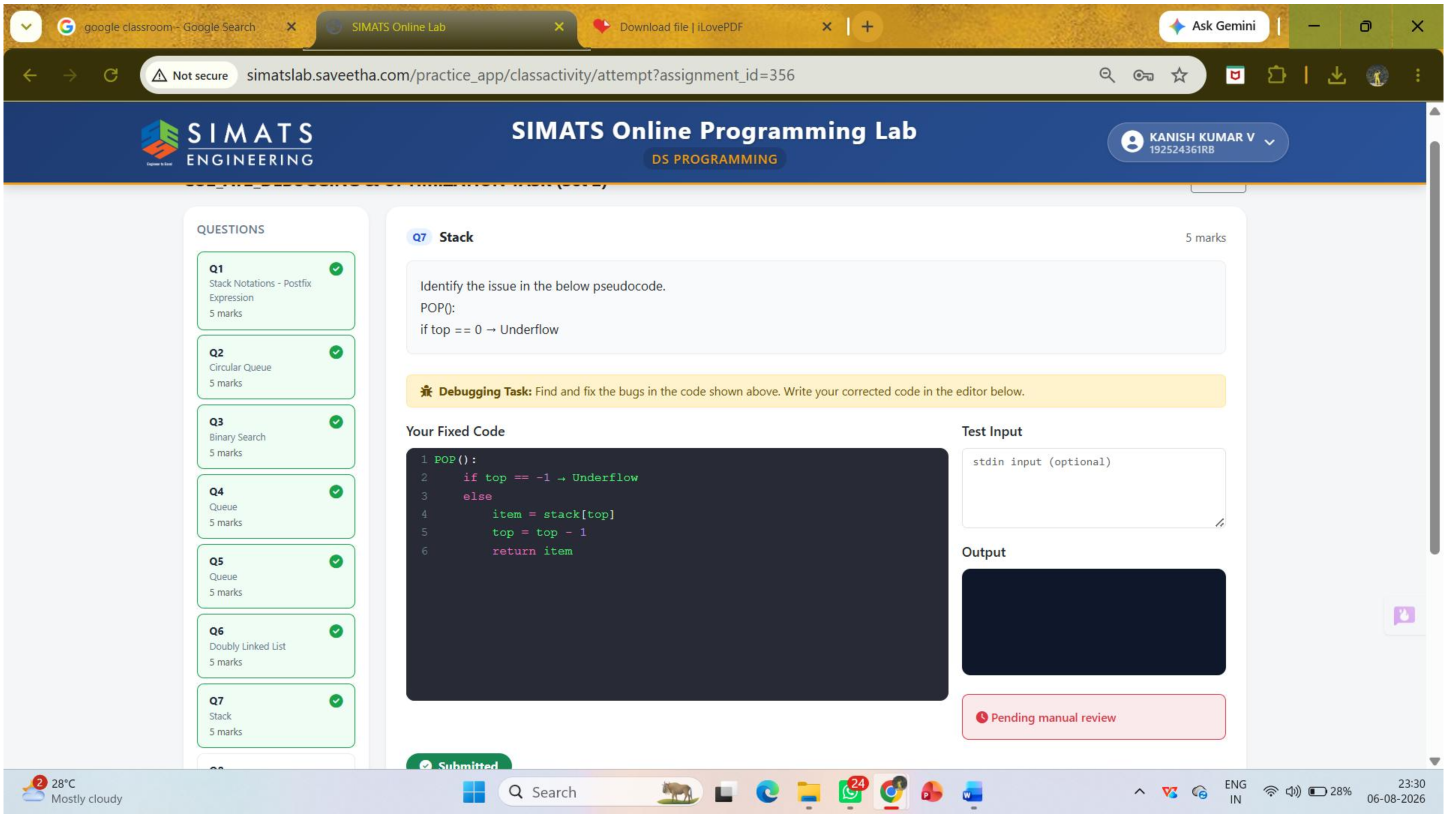

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QUESTIONS

Q1 Stack Notations - Postfix Expression 5 marks

Q2 Circular Queue 5 marks

Q3 Binary Search 5 marks

Q4 Queue 5 marks

Q5 Queue 5 marks

Q6 Doubly Linked List 5 marks

Q7 Stack 5 marks

Q7 Stack 5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
POP():
    if top == -1 → Underflow
    else
        item = stack[top]
        top = top - 1
        return item
```

Test Input

stdin input (optional)

Output

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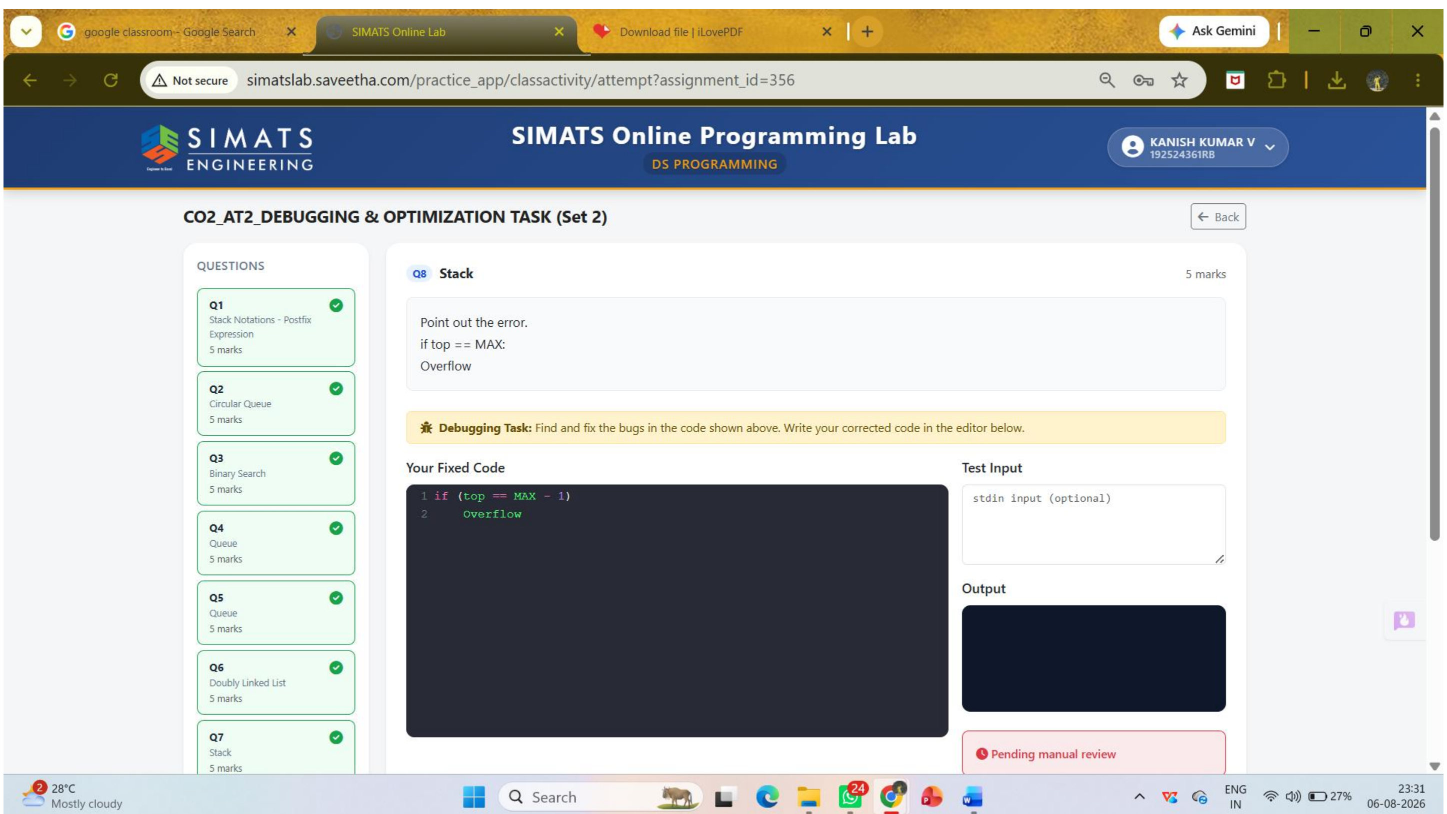
Submitted



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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1 Stack Notations - Postfix Expression 5 marks

Q2 Circular Queue 5 marks

Q3 Binary Search 5 marks

Q4 Queue 5 marks

Q5 Queue 5 marks

Q6 Doubly Linked List 5 marks

Q7 Stack 5 marks

Q8 Stack 5 marks

Point out the error.
if top == MAX:
Overflow

**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if (top == MAX - 1)
    Overflow
```

Test Input

stdin input (optional)

Output

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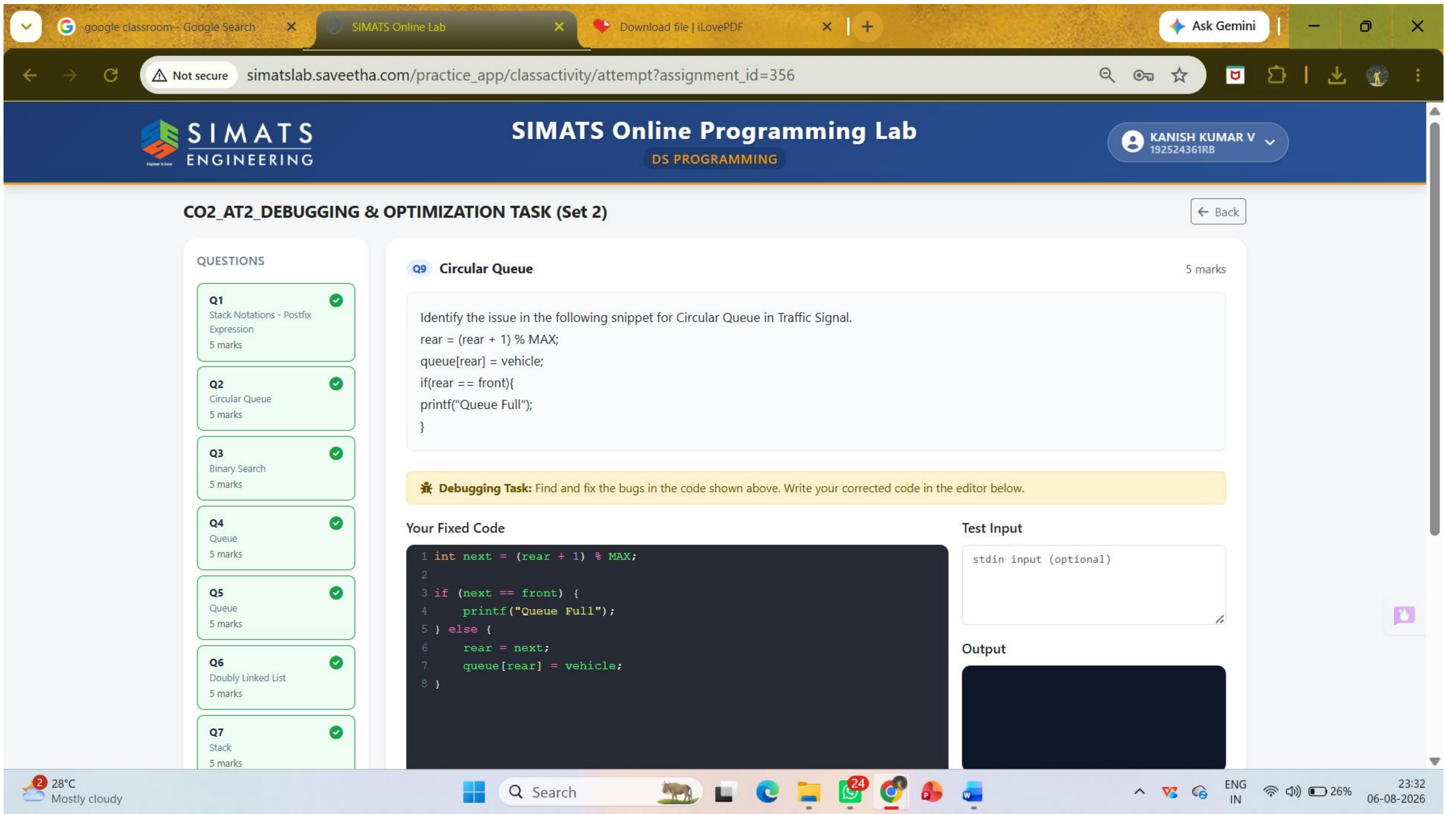

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

Q1 Stack Notations - Postfix Expression 5 marks
Q2 Circular Queue 5 marks
Q3 Binary Search 5 marks
Q4 Queue 5 marks
Q5 Queue 5 marks
Q6 Doubly Linked List 5 marks
Q7 Stack 5 marks

Q9 Circular Queue — 5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
int next = (rear + 1) % MAX;

if (next == front) {
    printf("Queue Full");
} else {
    rear = next;
    queue[rear] = vehicle;
}
```

Test Input

stdin input (optional)

Output

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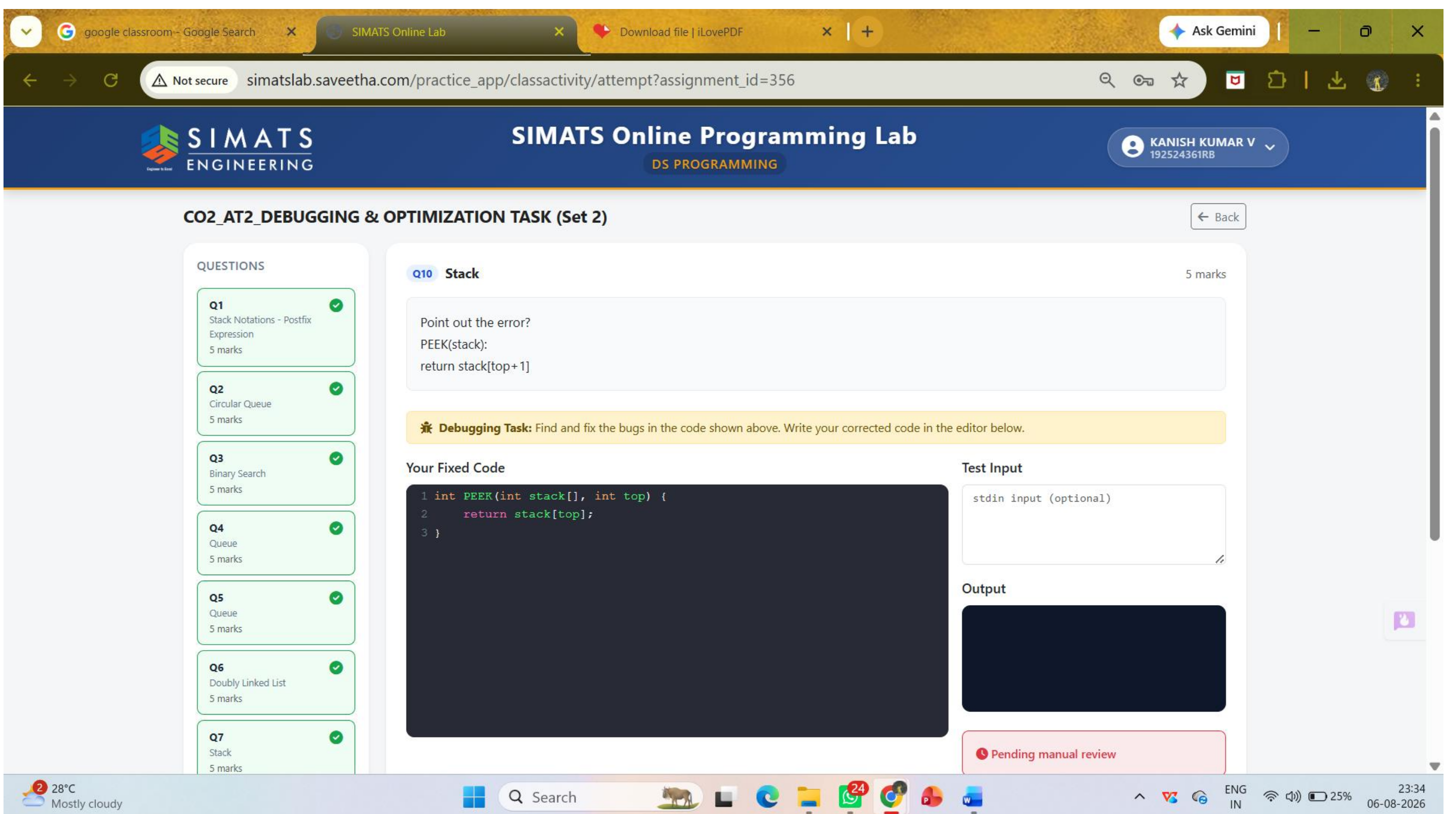

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

KANISH KUMAR V 192524361RB

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

Q1 Stack Notations - Postfix Expression 5 marks
Q2 Circular Queue 5 marks
Q3 Binary Search 5 marks
Q4 Queue 5 marks
Q5 Queue 5 marks
Q6 Doubly Linked List 5 marks
Q7 Stack 5 marks

Q10 Stack — 5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
int PEEK(int stack[], int top) {
    return stack[top];
}
```

Test Input

stdin input (optional)

Output

Pending manual review

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